

Stat 3113 Formula Sheet (Exam 1: Lectures 0-13)

Descriptive Statistics

- Sample Mean: $\bar{x} = \frac{1}{n} \sum x_i$
- Sample Variance: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$
- Sample Std. Dev.: $s = \sqrt{s^2}$
- IQR: $Q_3 - Q_1$
- Outlier Rule: $< Q_1 - 1.5(\text{IQR})$ or $> Q_3 + 1.5(\text{IQR})$.

General Probability Rules

- **Axioms:** $P(A) \geq 0$; $P(S) = 1$; For disjoint A, B, $P(A \cup B) = P(A) + P(B)$.
- **Complement Rule:** $P(A^c) = 1 - P(A)$
- **Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Conditional Probability:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- **Multiplication Rule:** $P(A \cap B) = P(A|B)P(B)$
- **Independence:** A and B are independent if $P(A|B) = P(A)$. For independent events, $P(A \cap B) = P(A)P(B)$.
- **Total Probability:** $P(B) = \sum P(B|A_i)P(A_i)$
- **Bayes' Theorem:** $P(A_j|B) = \frac{P(B|A_j)P(A_j)}{P(B)}$

General Discrete Random Variables

- **pmf:** $p(x) = P(X = x)$. Properties: $p(x) \geq 0$, $\sum p(x) = 1$.
- **cdf:** $F(x) = P(X \leq x) = \sum_{y \leq x} p(y)$
- **Mean (Expected Value):** $\mu = E[X] = \sum x \cdot p(x)$
- **Variance:** $\sigma^2 = V[X] = E[X^2] - \mu^2$

– where $E[X^2] = \sum x^2 \cdot p(x)$

- **Linear Transformations:** (for constants a, b)

– $E[aX + b] = aE[X] + b$

– $V[aX + b] = a^2V[X]$

Named Discrete Distributions

Binomial Distribution: $X \sim \text{Bin}(n, p)$

What it models: The number of "successes" (X) in a fixed number of trials (n).

Conditions:

1. A fixed number of trials, n .
2. Each trial is independent.
3. Each trial has two outcomes: Success or Failure.
4. The probability of success, p , is constant for all trials.

Probability Mass Function (pmf):

$$P(X = x) = \underbrace{\binom{n}{x}}_{\text{Number of ways to arrange } x \text{ successes}} \cdot \underbrace{p^x}_{\text{Probability of } x \text{ successes}} \cdot \underbrace{(1-p)^{n-x}}_{\text{Probability of } n-x \text{ failures}}$$

Mean & Variance:

- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1-p)$

Poisson Distribution: $X \sim \text{Poisson}(\mu)$

What it models: The number of events (X) occurring in a fixed interval of time, area, or volume.

Key Parameter:

- μ is the average number of events in that interval. It is a rate.

Conditions:

1. Events occur independently.
2. The average rate (μ) at which events occur is constant.
3. Two events cannot occur at exactly the same instant.

Probability Mass Function (pmf):

$$P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$$

Mean & Variance: A key property of the Poisson distribution.

- Mean: $E[X] = \mu$
- Variance: $V[X] = \mu$

Poisson Approximation to the Binomial

If a process is Binomial with a large number of trials n and a small probability of success p , its probabilities can be approximated by a Poisson distribution with mean $\mu = np$.

$$\text{Bin}(n, p) \approx \text{Poisson}(\mu = np)$$

Stat 3113 Definitions & Concepts

Study Design

Population: Entire group studied
Sample: Subset of population
Parameter: Population measure
Statistic: Sample measure
Observational Study: Observe only
Experiment: Apply treatments
Confounding: Effects indistinguishable
Control: Use control groups
Randomization: Random assignment
Replication: Sufficient sample size

Variables

Qualitative (Categorical):

- **Nominal:** No order (color, gender)
- **Ordinal:** Natural order (grades)

Quantitative (Numerical):

- **Discrete:** Countable (no. children)
- **Continuous:** Any value (height)

Descriptive Stats

Center:

- **Mean:** Average, not resistant
- **Median:** Middle value, resistant
- **Mode:** Most frequent

Spread:

- **Range:** Max - Min
- **IQR:** $Q_3 - Q_1$, resistant
- **Std Dev:** Typical distance from mean
- **Variance:** $(StdDev)^2$

Shape:

- **Symmetric:** Mean $\approx$ Median
- **Left Skewed:** Mean $<$ Median
- **Right Skewed:** Mean $>$ Median

Resistant: Unaffected by outliers (median, IQR)

Non-resistant: Affected by outliers (mean, std dev)

Probability

Sample Space: All possible outcomes
Event: Subset of sample space
Probability: Value between 0 and 1
Conditional Prob: $P(A|B)$ - prob of A given B
Independence: $P(A|B) = P(A)$
 $\Rightarrow P(A \cap B) = P(A)P(B)$
Mutually Exclusive: $P(A \cap B) = 0$
Complement: $P(A^c) = 1 - P(A)$

Random Variables

Random Variable: Assigns numbers to outcomes
pmf: Prob mass function (discrete)
 $p(x) = P(X = x)$
pdf: Prob density function (continuous)
Note: $P(X = c) = 0$ for continuous RV
cdf: $F(x) = P(X \leq x)$
Expected Value: Long-run average
Variance: Measures spread
 $\text{Var}(X) = E[(X - \mu)^2]$

Distribution Properties

Binomial: Fixed trials, constant prob
Poisson: Events in fixed interval
Normal: Bell-shaped, symmetric
Standard Normal: $\mu = 0, \sigma = 1$
Uniform: All values equally likely
Exponential: Waiting times, memoryless